

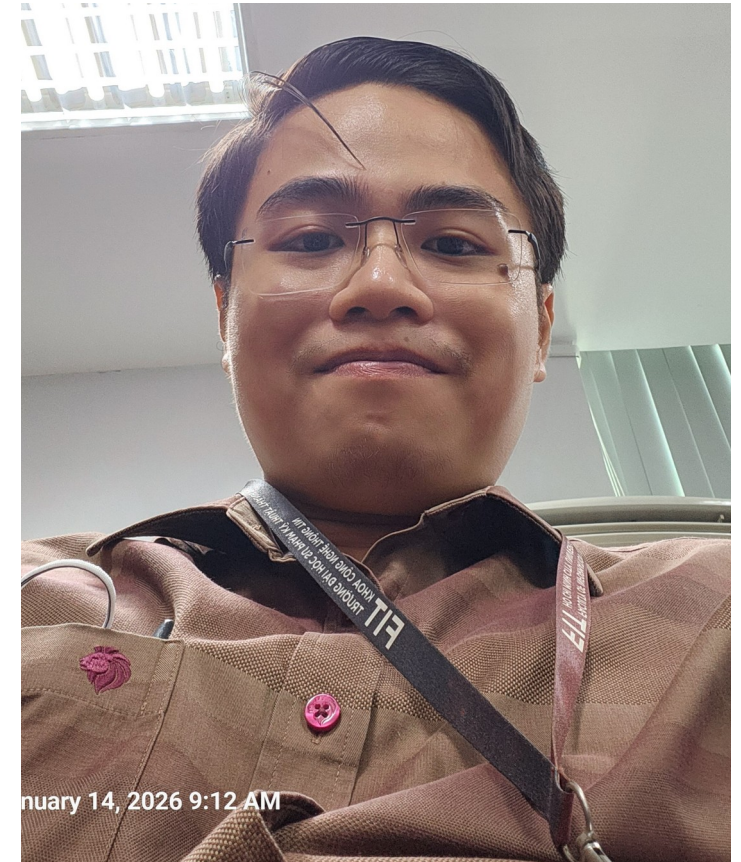
Day 1

How data science works

Lecturer: Msc. Minh Tan Le

I. Introduction

- Msc. Minh Tan Le
- Lecturer of FIT's Artificial Intelligence
- Fields:
 - Spatio-temporal models
 - Generative AI
 - Face Analysis
 - Society & Education



Goals

- I. A few concepts
- II. How we communicate
- III. How we work
- IV. Materials

Data

Science

Experiments

Python/R

Processing

Insights

Data science

Data engineer

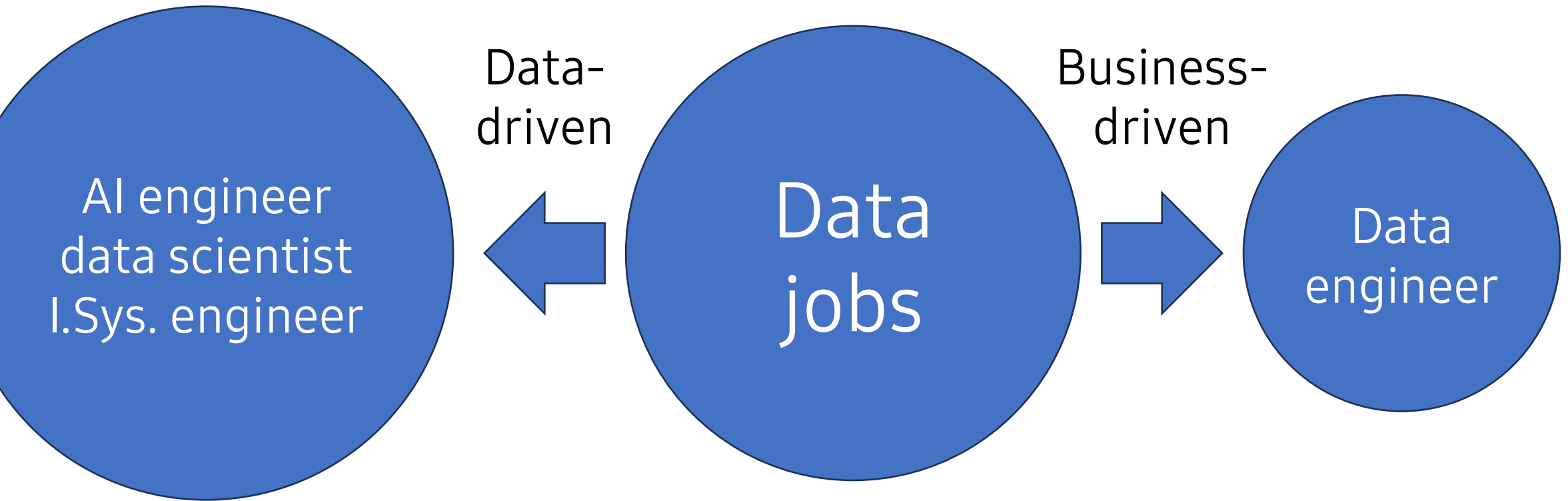
Definition

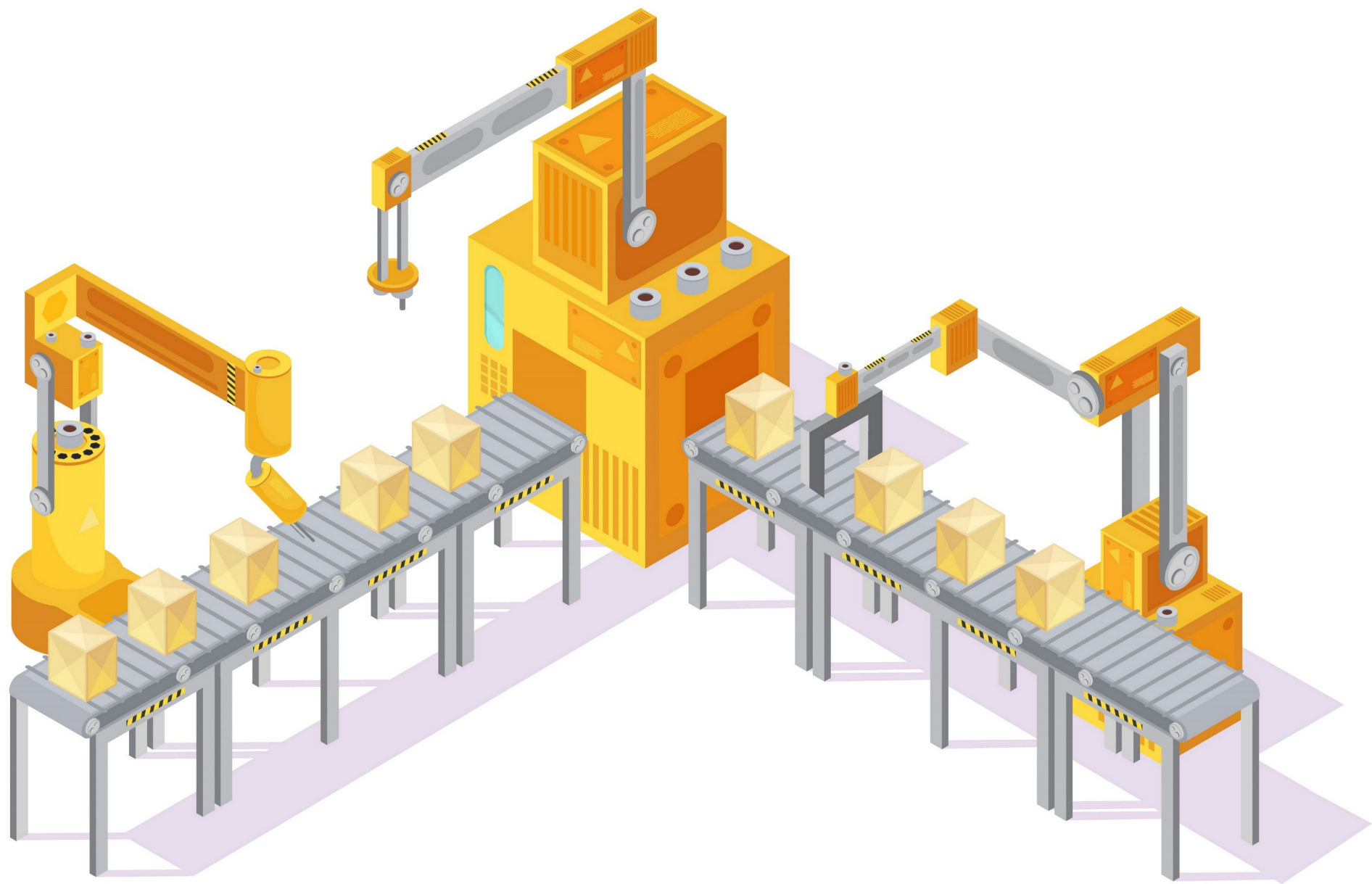
Data science

- Combine methods, principles from statistics & computer science to **work with and draw insights from data.**

Data engineer

- Design, build a system **to collect and analyze raw data** from multiple sources and formats.







Apache
Airflow



Batch layer

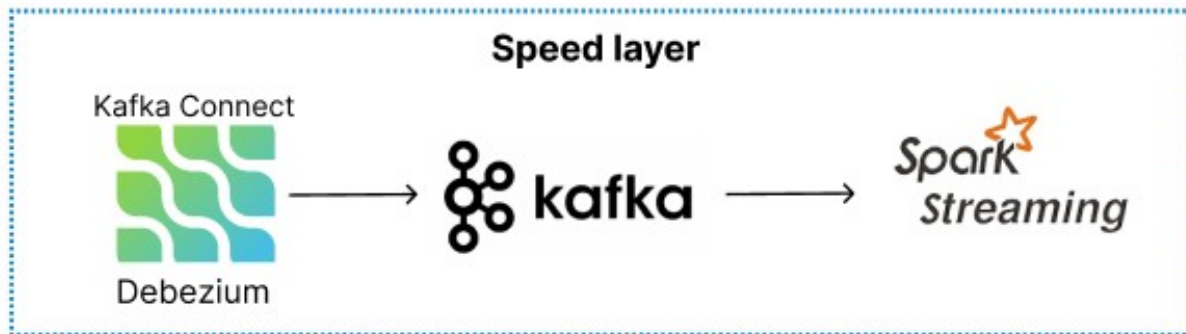
APACHE
Spark



Datalake

APACHE
Spark

Data source



Speed layer

Kafka Connect



Debezium



kafka

Spark
Streaming

Serving layer

Batch Views



Speed Views

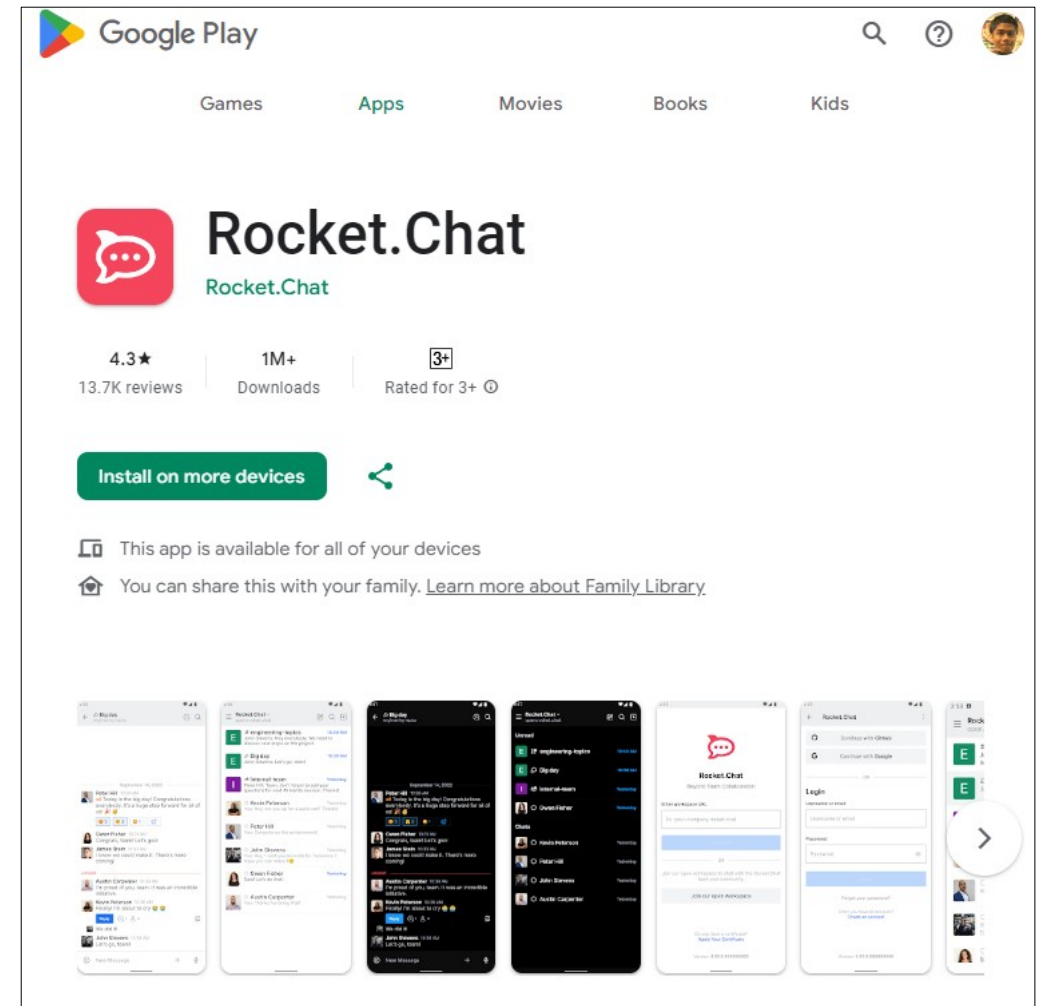
Anyway, when working with data, you must...

- Love (or at least not hate) the data.
- Creative.
- Stay in context.
- Open-minded.

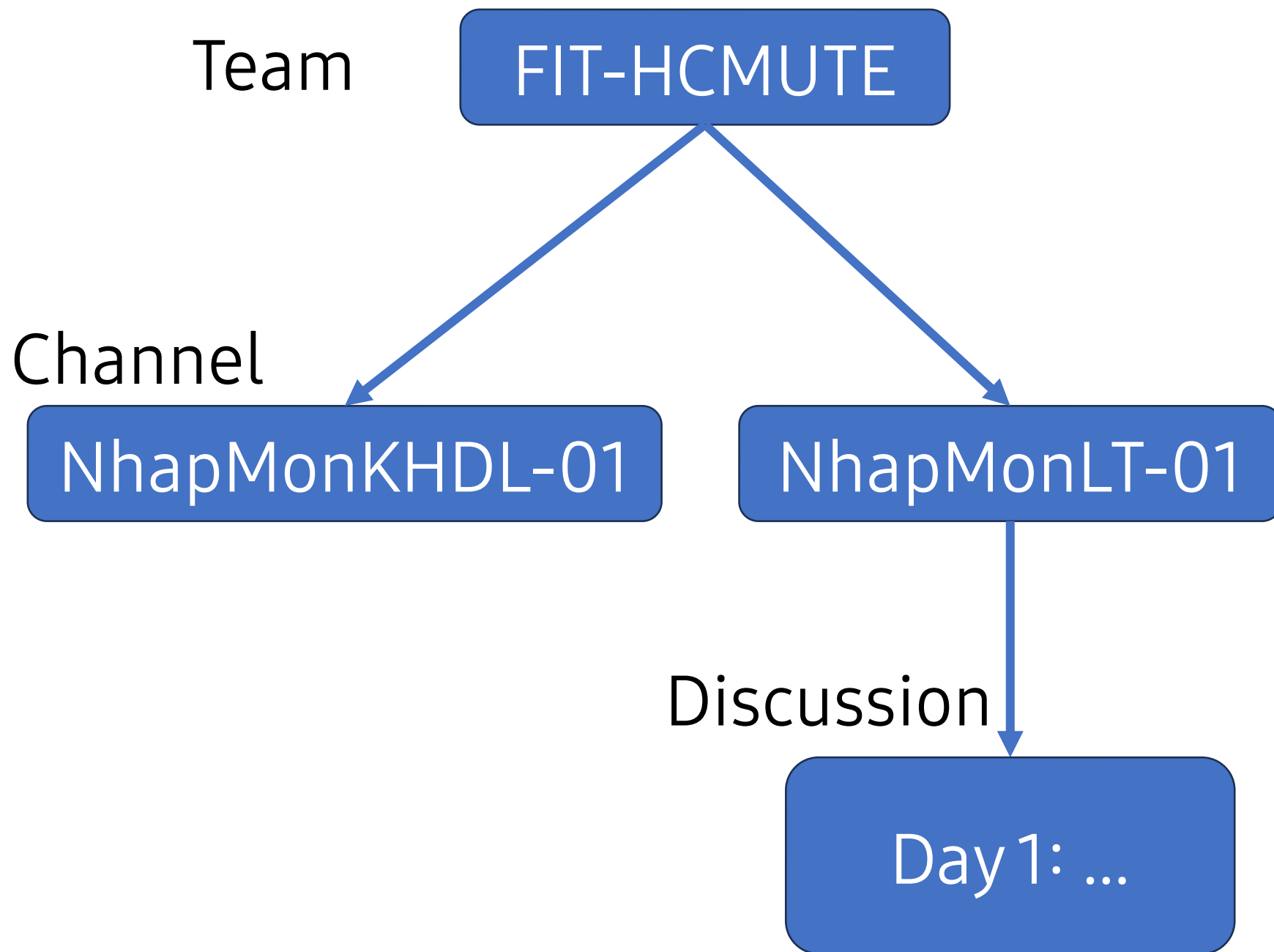
II. How we communicate

- By chatting platform:
 - **Website:** <https://chat.cntt.io>
 - **Android/iOS:** Rocket.Chat
 - **Desktop:**

<https://github.com/RocketChat/Rocket.Chat.Electron/releases>



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How we work

- Form groups of **2-3 students**.
- Vote for a leader.
- Go to IO chat and send your group info in the discussion:
 - Members' name and student ID.
 - Who's the leader?

O'REILLY®

Learning Data Science

Data Wrangling, Exploration, Visualization,
and Modeling with Python



Sam Lau,
Joseph Gonzalez
& Deborah Nolan

Here's the plan

- Credits: 3
- Works
 - Homework/Exercises (X1)
 - Midterm (X3): Group presentation
 - Final: Group face-to-face report

Midterm

- Outputs
 - Short report (docx, 10-20 A4 pages)
 - Slides (pptx, 15 mins)
- Tasks:
 - Imagine: What's your big, awesome project.
 - Choose at least 3 datasets.
 - Introduce the datasets: Samples, fields, statistics, qualities.
 - **General** comparison & conclusion.

Final

- Output
 - Report (30 – 60 A4 pages)
 - Source codes
- Tasks
 - Everything we did in midterm
 - Project scopes
 - Data insights
 - **deep** comparison & conclusion
 - 1-page Report & 1-page comparison report

Midterm: Notes

- Each group picks...
 - A project & 3 datasets that are **unique**.
 - The datasets must be **downloadable & interchangeable**.

Midterm

- Outputs

- Short report (docx, 10-20 A4 pages)
- Slides (pptx, 15 mins)

- Tasks:

- Imagine: What's your big, awesome project.
- Choose at least 3 datasets.
- Introduce the datasets: Samples, fields, statistics, qualities.
- Comparison & conclusion.

This week's homework!

Ask a
question



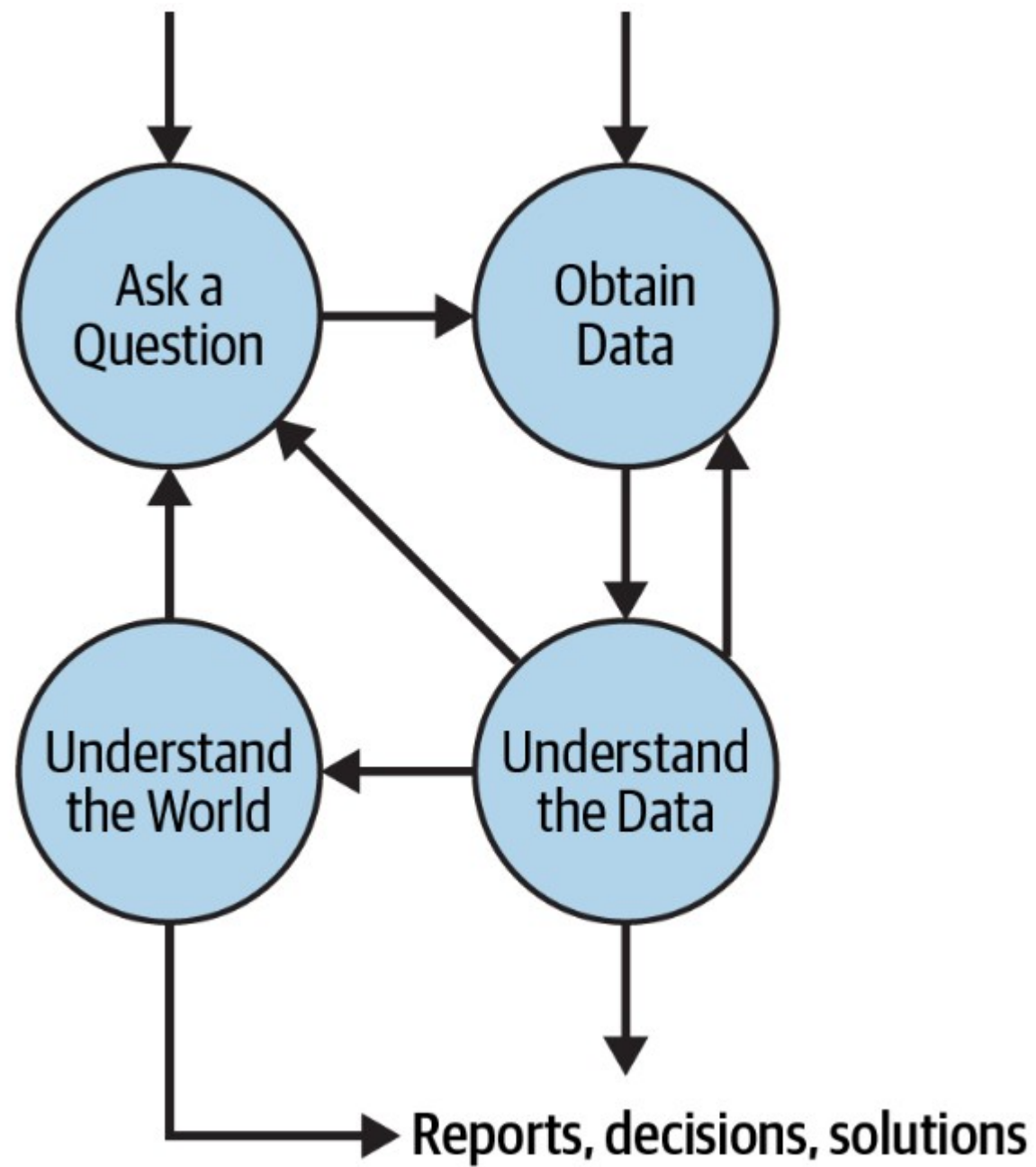
Obtain data

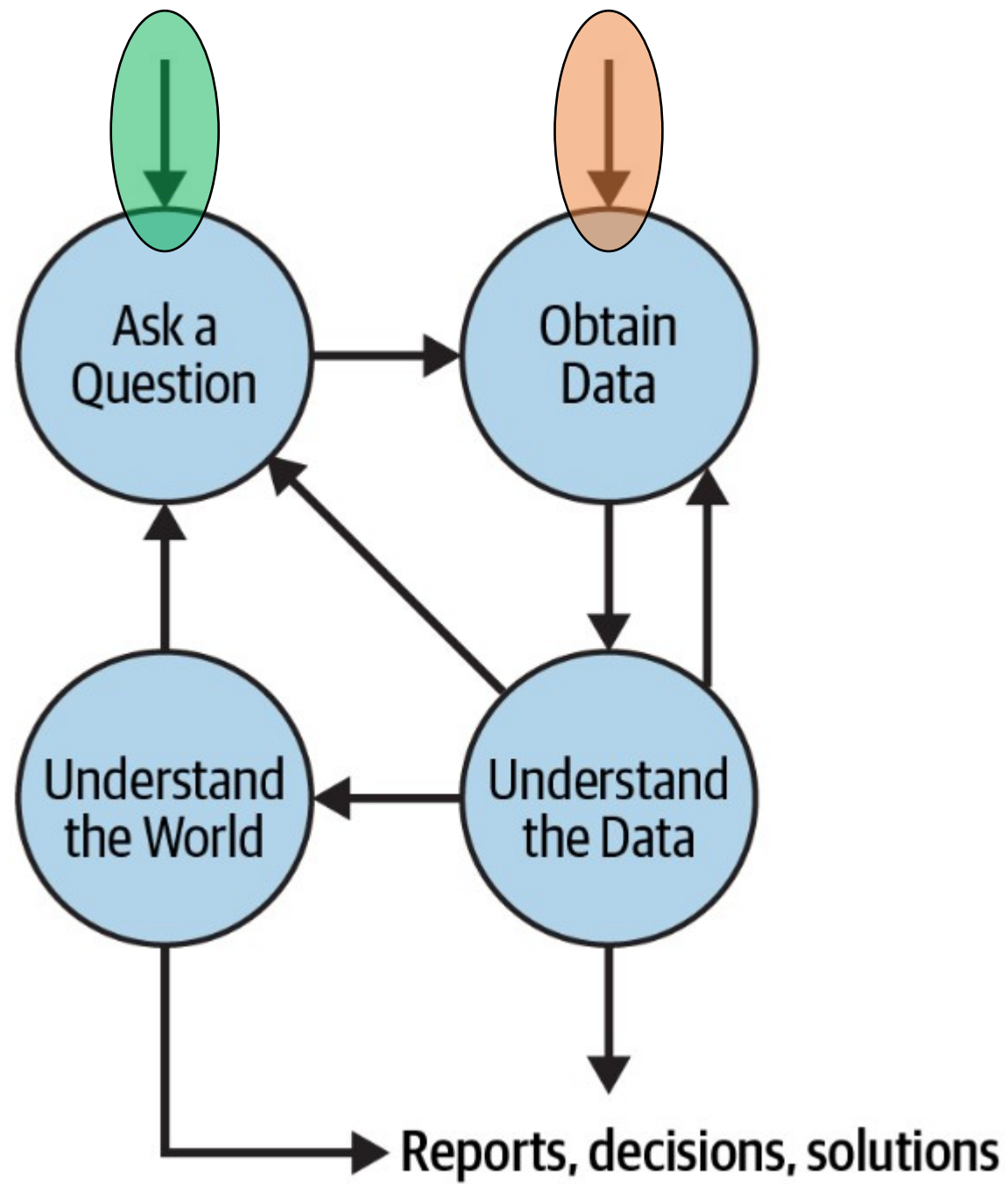


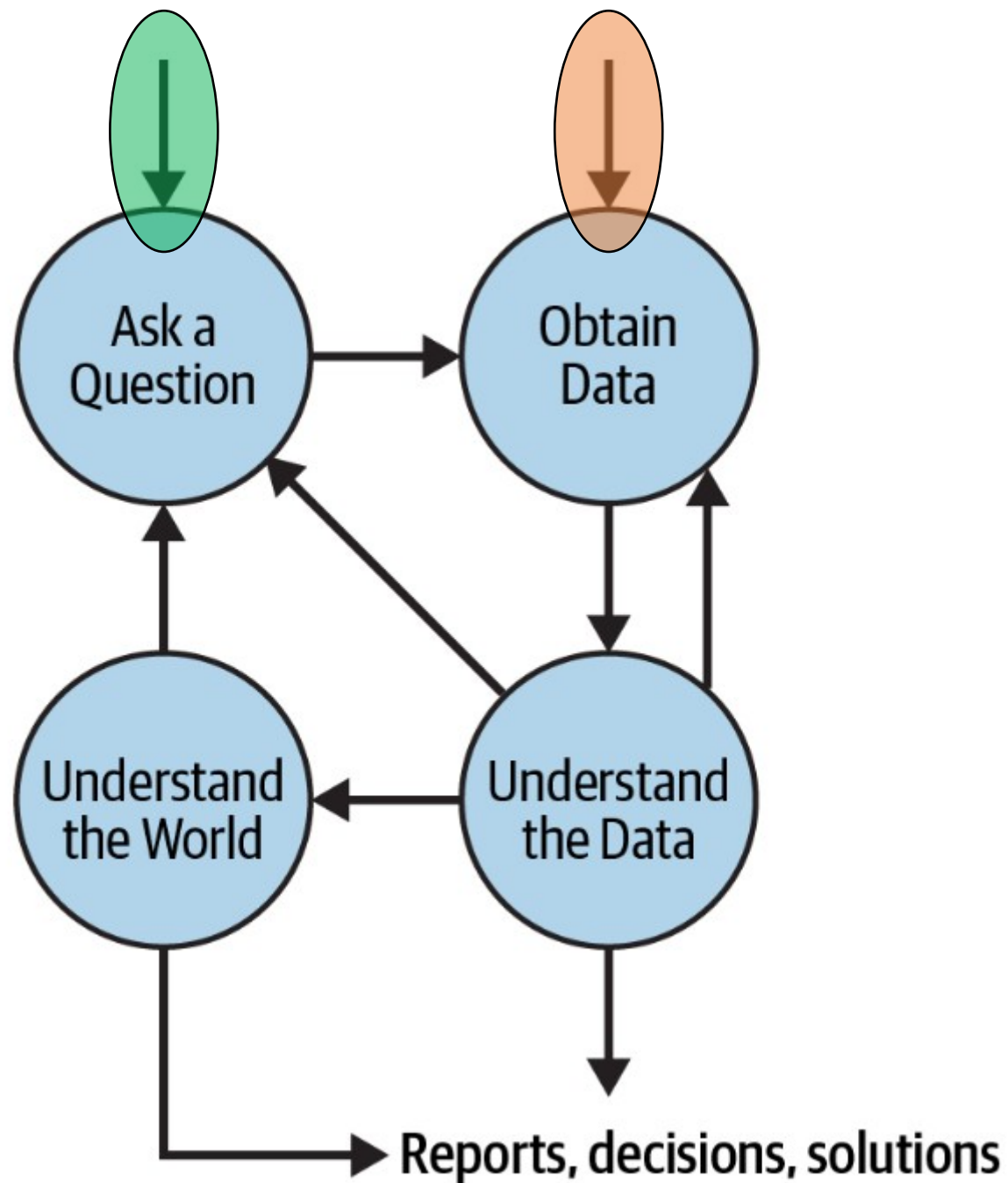
Understand
the data



Understand
the world







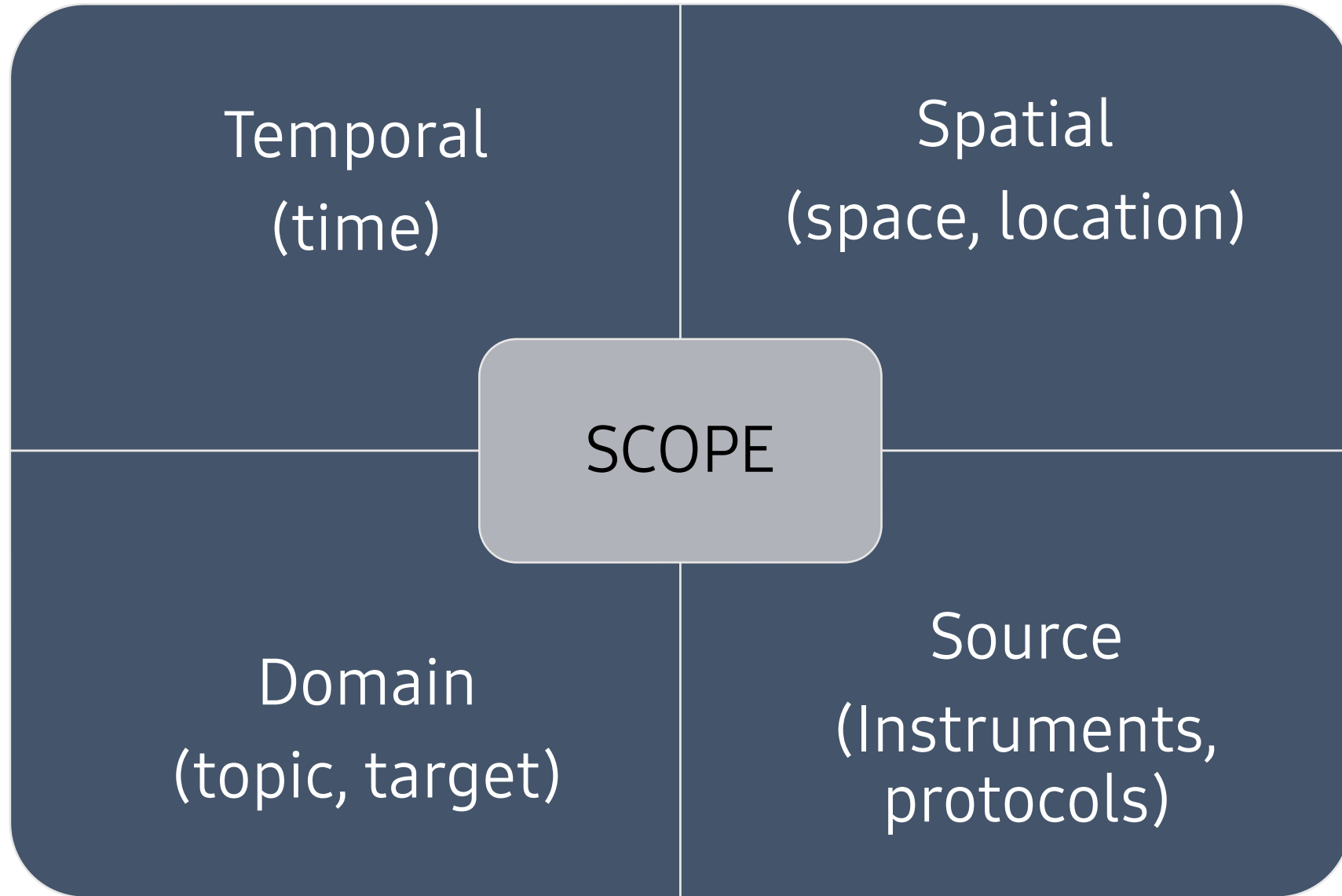
Scene #1: Your customer wants to improve their B2C e-commerce website by blocking spam, harassment comments.

Scene #2: Your customer wants you to build a model that recommends suitable insurance products.

	Suburb	Address	Landsize	YearBuilt
0	Abbotsford	85 Turner St	202	
1	Airport West	154 Halsey Rd	303	2016
2	Albert Park	105 Kerferd Rd	120	1900
...	...	...	...	...

We want our data to be **useful**, but...

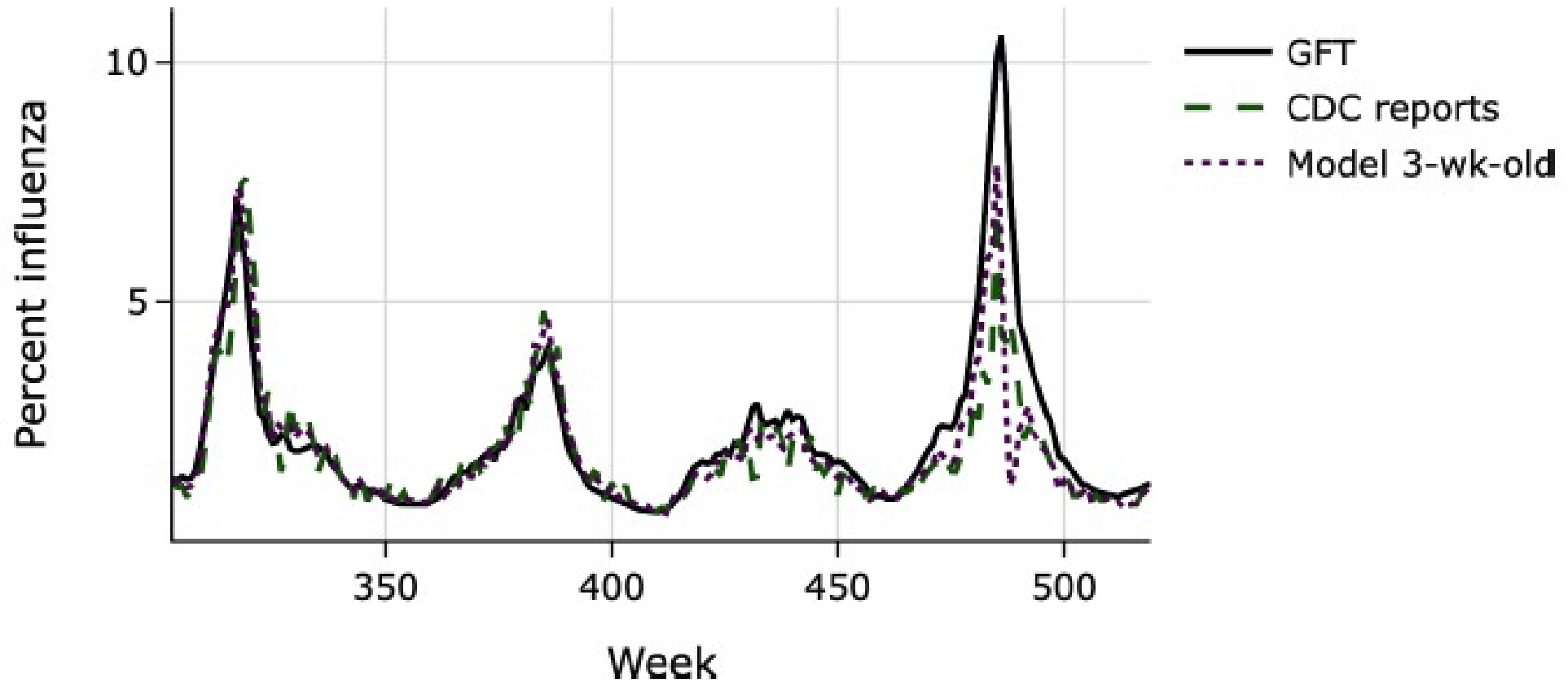
- It always has limitations.
 - Scope
 - The number of data collectors
 - Not all fields are reachable
 - Not all data are usable



Use case: Google Flu Trends (GFT)

- Since: 2007, abandoned in 2015.
- Base on user searches that are flu-related.
- Ideas: Counting can estimate the number of flu cases.

So what went wrong?



Concepts

- Target population (unit/atom): $\mathcal{P}$
- Access frame: Accessible elements for measurement & observation - $\mathcal{A}$
 - Can be a subset of target population
- Sample: The subset of $\mathcal{A}$ that is used to observe & measure.

Differences that can affect the results

- Time
 - How fast the data will be outdated (or it's already outdated)
 - Can it be temporarily outdated
- Place
 - Culture, population, environment, legality

Scope define how we collect data

- Instruments: Answer what can be used for collecting.
- Protocols: Answer how we make use of instruments.
- Instruments & Protocols are related.

Homework

- A docx which includes:
 - Project name
 - Project short description, plans (10-30 sentences)
 - Chosen datasets
 - Name
 - Author(s)
 - Downloadable links

Project short description (10-30 sentences)

- What is it about?
- Project scope: Where & when will it take place? For who?
- Data collection: What are the P, A, S? The factors that can affect the usefulness? The instruments & protocols?
- ...

Homework scores (1 week)

- Project (4 pts)
 - Name: 1 pt
 - Short description quality: 3pts
- Datasets (6pts, 2 pts each)
- **Non-interchangeable (same scope): -3 pts**
- Total fields < 10: -2 pts
- At least 500 records for largest table: -1 pt